

**ADMINISTRATIVE INFORMATION****FMI No: 25395****FMI Title:** VCT Global Table Offset Correction

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☒ Mandatory Safety	GEMS - AM:	Issue Date:
☐ Mandatory	GEMS - E:	
☐ Mandatory on Request	GEMS - A:	Close Date:

☒ Staged FMI	Customer List provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.
☒	Parts and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.
☐	Parts Ordering: GEMS - Americas, Contact your Region logistics coordinator GEMS - Europe, Order FMI kits from GPO/EPL Buc GEMS - Asia, Contact your region logistics coordinator
☐	No Parts are required to complete this FMI.
☐ Batch with	To minimize costs, combine implementation of this FMI with the FMIs listed for all customers where the effectivity is the same (Please see your customer lists).
☐ Non-Staged	No Customer List is provided. Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

Technical Support

GEMS - Americas	GEMS - Europe	GEMS – Asia
GE Medical Systems On-line Center 800-321-7937 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:

Sites running **06MW29.7** software require the **Limited Access Software CD (P/N 5192389)** to be loaded prior to loading VCT Global Table Correction Patch. Most systems will have this, but if yours does not, see Appendix A to order from Coakley

Unit Tracked: Refer to Effectivity list on page 2.**FMI Instructions Part No.:** 5197220-100**Do NOT leave unsecured on site**

FMI Completion Certification Report

Mail To

GEMS - Americas

G.E. Medical Systems
Attn: FMI Admin. W-523
P.O. Box 414
Milwaukee, WI 53201

GEMS - Europe

GEMS - Europe
FMI Admin. 322
BP34
F78533 Buc Cedex, France

Send to your
Country FMI or Service
Administrator

FMI No.: 25395

FMI Title: VCT Global Table Offset Correction

Customer Full Name:

Number and Street Address:

City and Country:

Country Code

Site Number

System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not in area. Product Locator notified

Customer Refusal or Equipment Altered - Modification Pending.

Installer

Comments

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2. Revision History

Revision	Date	Reason for Change
1	1/30/2007	Initial release.
2	2/05/2007	Added Notes in Section 11 & 12 on what will happen if the patch fails to install due to prerequisites for FMI not being met.
3	3/15/2007	Add Flash Download Patch and Instructions to Doc

3. Effectivity

This FMI applies to all LightSpeed 7.x (VCT64, VCT32, and VCT Select) Systems with the following Gantry model numbers:

Product Line	System	Gantry Model	Shipped Before
LightSpeed 7.x	VCT 64	5124069	Feb. 2, 2007
	VCT 32/Select	5126093	Jan. 15, 2007

4. Purpose

It was found that a potential inaccuracy can occur in the positional display when the manual cradle release on VT 1700 and VT 2000 patient tables is used inappropriately during Radiation Therapy simulation procedures.

The manual cradle release feature is intended for emergency egress and small movements during CT fluoroscopy procedures. In rare instances, if the cradle release is used to move a patient out of the gantry, instead of using the table control buttons as specified in the operator manual, a discrepancy of up to 6mm between the numerical display on the gantry and the actual position of the table can be introduced.

This addresses the following PSRs:

- **RT16/Xtra: 5mm offset table issue in Mayo** (13087519)
- **RT16: 5mm offset table issue in Mayo for radiotherapy** (13087764)

5. Prerequisite FMIs

Sites with 06MW03.4 SW : FMI 25383 – VCT M4 06MW03.4 PATCH A

Sites with 06MW29.7 SW : NONE

6. *Related FMIs*

FMI 22914 – RT16 GT Offset Correction.

7. *Furnished Materials*

FMI Kit 5197220 consists of the following materials:

Quantity	Description	Part #
1	FMI document	5197220-100
1	VCT/VCT Select GT Patch for 06MW29.7	5196620
1	VCT/VCT Select GT Patch for 06MW03.4	5196853
1	TCB Flash Download Patch	5173947
1	VCT Tips and Workarounds-Multi-Language	5116420-247
1	LightSpeed 7.X Applications Tips and Workarounds – English	5116420-100
1	LightSpeed 7.X Applications Tips and Workarounds – French	5116420-101
1	LightSpeed 7.X Applications Tips and Workarounds – Spanish	5116420-106
1	LightSpeed 7.X Applications Tips and Workarounds – German	5116420-108
1	LightSpeed 7.X Applications Tips and Workarounds – Italian	5116420-111
1	LightSpeed 7.X Applications Tips and Workarounds – Portuguese	5116420-127
1	LightSpeed 7.X Applications Tips and Workarounds – Japanese	5116420-140
1	LightSpeed 7.X Applications Tips and Workarounds – Chinese	5116420-141

8. *Tool Requirements*

No special tools required.

9. *Personnel Requirements*

Resource	1 FE
Labor	0.5 hrs
Travel	1 hr

10. Procedure

Confirm System Software Configuration

1. Open a Unix shell as ctuser.
2. Type **swhwinfo -all <ENTER>**
3. The returning string should include the following software revision:

For VCT M4 Systems: 06MW03.4

- Proceed to Section 11 – 06MW03.4 Install Procedure

For VCT Limited Access Systems: 06MW29.7

- Proceed to Section 12 – 06MW29.7 Install Procedure

11. 06MW03.4 Install Procedure

Verify Prerequisite Software Installed

1. Open a Unix shell as ctuser.
Type **showprods | grep -i patch <ENTER>**
The output should contain the two following patches:
VCT_M4_Flash_Download_patch-1-1
VCTM4_patchA

The TCB Flash Download Patch (included in this kit) and VCT Patch A are both required before loading the software from this FMI. If the two patches are NOT part of the output, get the patches from the following sources and load prior to loading VCT GT Offset Correction Software.

- **P/N 5173947 – TCB Flash Download Patch for 06MW03.4 VCT M4** provided in this kit. See Appendix D for install instructions.
- Order **P/N 5179392 – VCT M4 06MW03.4 PATCH A** from Coakley, or download from CT Online Center Support Central Website. See Appendix A for links.

Install VCT GT Offset Correction Software

1. Shutdown Applications.
2. Install patch CD p/n 5196853 into HOST computer.
 - a. Type: **su - <ENTER>**
 - b. Type: **patch_install -c <ENTER>**
 - c. Type: **y <ENTER>**The load will take about 30 seconds and the following output will be displayed:
M4_GT_5mm_Offset_Patch has successfully been installed.
3. Verify software loaded correctly.
 - a. Type **showprods | grep -i patch <ENTER>**
The output should display:
VCT_M4_Flash_Download_patch-1-1
VCTM4_patchA
M4_GT_5mm_Offset_Patch-1-1
4. Eject CD, type **eject <ENTER>**
5. Type: **reboot <ENTER>**

NOTE: If a pink box pops up during the boot process stating, “install_patch failed / uninstall_patch failed,” refer to *Verify Prerequisite Software Installed* under Section 11, 06MW03.4 Install Procedure. You do not have all of your prerequisites installed.

6. Run Flash download tool (Query and Update) located on the Common Service Desktop under Utilities.
7. Perform **Verify Patch Functionality** in Section 15, Appendix C.
8. Perform a test scan to verify system functionality.
9. FMI Completed. Remove instructions from site and close dispatch per normal regional procedure.

12. 06MW29.7 Install Procedure

Verify Prerequisite Software Installed

1. Open a Unix shell as ctuser.
2. Type: **showprods | grep -i patch <ENTER>**

The output should contain:

Limited_Access_patch

The Limited Access software patch is required to be installed before loading the software from this FMI. If this is **NOT** part of the output, order the following patch from Coakley and install prior to loading VCT GT 5MM Offset Correction Software:

- P/N 5192389 – LIMITED ACCESS SOFTWARE CD

NOTE: For detailed instructions on how to order software CDs from Coakley, see Appendix A.

Install VCT GT 5mm Offset Correction Software

1. Shutdown Applications.
2. Open a Unix shell as ctuser.
3. Type: **touch /usr/g/fw/tgpFtp /usr/g/fw/orpFtp <ENTER>**
4. Type: **cd /usr/g/fw <ENTER>**
5. Type: **ls *Ftp <ENTER>**
Verify that **tgpFtp** and **orpFtp** exist. (If those two files are not found or misspelled, repeat the step 3.)
6. Install patch CD p/n 5196620 in the Host computer.
 - a. Type: **su - <ENTER>**
 - b. Type: **patch_install -c <ENTER>**
 - c. Type: **y <ENTER>**The load will take about 30 seconds and the following output will be displayed:
JALO_GT_5mm_Offset_Patch has been successfully installed.
7. Verify software loaded correctly.
 - a. Type: **showprods | grep -i patch <ENTER>**The output should be:
Limited_Access_patch

JALO_GT_5mm_Offset_Patch-1-1

8. Eject CD, type **eject** <ENTER>
9. Type: **reboot** <ENTER>

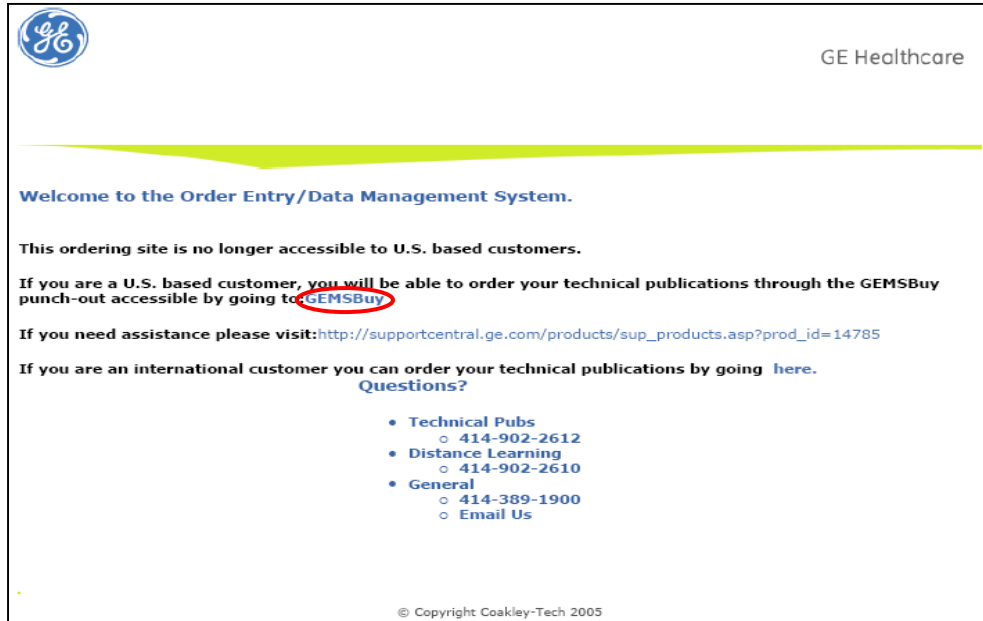
NOTE: If a pink box pops up during the boot process stating, “install_patch failed / uninstall_patch failed,” refer to *Verify Prerequisite Software Installed* under **Section 12, 06MW29.7 Install Procedure**. **You do not have all of your prerequisites installed.**

10. Run Flash download tool (Query and Update) located on the Common Service Desktop under Utilities.
11. Perform **Verify Patch Functionality** in Section 15, Appendix C.
12. Perform a test scan to verify system functionality.
13. FMI Completed. Remove instructions from site and close dispatch per normal regional procedure.

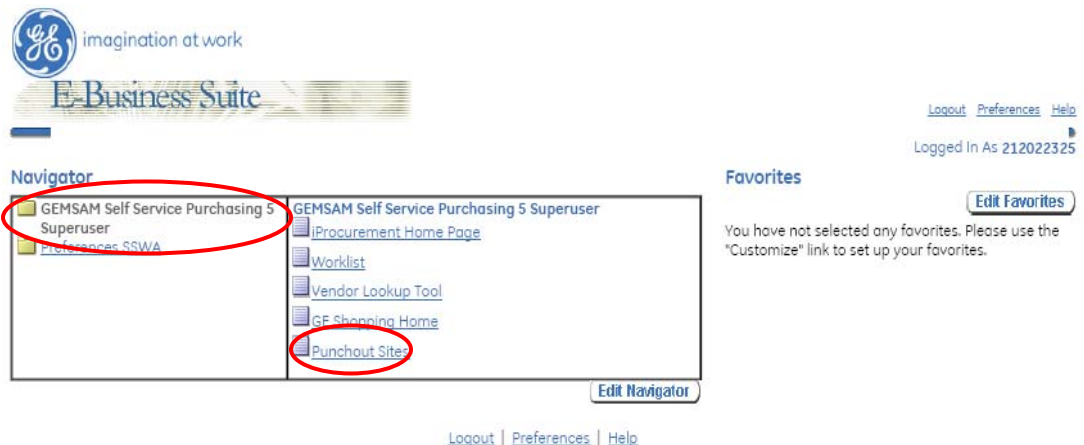
13. Appendix A

Coakley Website

1. Go to the following link: <http://coakleytech.med.ge.com/cgi-win/default.aspx>



2. Click on **GEMSBuy** link.
3. Click on **GEMSAM Self Service Purchasing 5 Superuser**.



4. Click on **Punchout Sites**.

5. Click **Coakley Tech**

Name	Description
AQ Safety	MRO - Prescription Safety Glasses
Barnes & Noble	Office - Books
Bearton Distributors Inc	MRO - Power Transmission
Coakley Tech	Office - Technical Publications
CSC	Office - Promotional Materials
Dell US	Tech - Laptops, Desktops and Accessories.
Ergonomix	Office - Ergonomic equipment
Evigga	Office - Promotional Items
EviggaOlympic	Office - Promotional Items
Fastenal	MRO - Fasteners and Industrial Supplies
Fine Arts	Office - Business Cards
Fisher Scientific U.S.A. Ltd.	MRO - All Laboratory Equipment, Consumables and Chemicals

6. Enter the CD P/N and submit the order.

CT Online Center Support Central Website

- Proceed to
http://supportcentral.ge.com/bookupload/sup_book_default.asp?prod_id=15768&trng_doc_id=281360&vendor_flag=11&doc_cat_id=115255.
- Select Folder **LightSpeed 7.x (VCT64, VCT32)**
- Links to Patch A Download, ISO File and Instruction.**
 - Link to Read-Me File for Patch A
http://supportcentral.ge.com/bookupload/sup_book_access.asp?book_id1=3155808&chap_id=3019134&book_id2=347853&doc_id1=281360&prod_id1=15768
 - Link to ISO File.
http://supportcentral.ge.com/bookupload/sup_book_access.asp?book_id1=3155810&chap_id=3019134&book_id2=347853&doc_id1=281360&prod_id1=15768
 - Link to FMI 25383 Instructions (Information regarding Patch A)
http://supportcentral.ge.com/bookupload/sup_book_access.asp?book_id1=3155805&chap_id=3019134&book_id2=347853&doc_id1=281360&prod_id1=15768

14. Appendix B

Remove / Uninstall GT_5mm_Offset_Patch-1-1:

1. Shutdown all applications
2. Open Unix Shell
3. Type **su -** <ENTER>
4. Type **patch_install -u** <ENTER>

CAUTION: The tool will ask to uninstall all patches installed on the system. Only type “Y” for **GT_5mm_Offset_Patch** to be uninstalled.

You will see either of the following message:

M4_GT_Patch-1-1 has been uninstalled successfully.

or

JALO_GT_Patch-1-1 has been uninstalled successfully.

7. Verify that prerequisite software is installed on system.
 - a. Type: **showprods | grep -i patch** <ENTER>
The output should contain the two following patches:
 - If software version is 06MW03.4:
VCT_M4_Flash_Download_patch-1-1
VCTM4_patchA
 - If software version is 06MW29.7:
Limited_Access_patch
8. Type **reboot** <ENTER>
Wait for applications to start.
9. Run Flash download tool (Query and Update) located on the Common Service Desktop under Utilities.

15. Appendix C

Verification of Patch Functionality

1. Move the table to home position.
 - a. Lift the table to the highest position and press “Cradle Out” to move the cradle backward to electrical limitation.
2. Unlatch the cradle and pull the cradle out backward to its mechanical limitation until the cradle can't be pulled anymore.
3. Press the “Landmark” button, since the downloaded firmware is disables the setting of the landmark is this special condition, you may observe that the landmark can not be set then.
4. The error log should be the following; the “Cradle position” and “IMS position” should refer to actual values and may not be the same as follows.

```
Exception Class : Secondary   Severity : Pri/Soft
File : uif_table_tilt_mgr_gt.c   Line# : 6057
Function : No System Function Reported
Scan Type : None/Unknown Scan: 0/0/0
Landmark button is ignored because it's pressed in invalid location.
Press <Cradle-IN> and press Landmark again.
Cradle position -4590.
IMS position 0.
```

16. Appendix D

TCB Flash Download Install Instructions

1. Insert the patch CD into the DVD Drive of the SCSI tower.
2. Shutdown application from common service desktop.
3. Open a Unix Shell.
 - a. Type: **su** – <ENTER>
 - b. Type: **#bigguy** <ENTER>
 - c. Type: **mount /mnt/cdrom1** <ENTER>
 - d. Type: **cd /mnt/cdrom1** <ENTER>
 - e. Type: **./install_patch -p VCT_M4_Flash_Download_patch-1-1.i386.rpm** <ENTER>
 (The following installation message appears.)


```
*****PRE-INSTALL*****
No pre-install activity required
*****PRE-INSTALL*****
*****INSTALL*****
Installing VCT_M4_Flash_Download_patch-1-1
##### [100%]
##### [100%]
*****INSTALL*****
*****POST-INSTALL*****
No post-install activity required
*****POST-INSTALL*****
```
 - f. Type: **cd** <ENTER>
 - g. Type: **umount /mnt/cdrom1** <ENTER>
4. Eject and remove the patch CD from the DVD drive of SCSI tower.
5. Verify the Flash Download patch is installed:
 - a. Type: **rpm -qa|grep patch** <ENTER>
 (The output should be similar to the following)
VCT_M4_Flash_Download_patch-1-1
6. Type the following to exit from Unix shell:
 - a. **exit** <ENTER>
 - b. **exit** <ENTER>